LEARNING MODULE

12

Cholesky, SVD and Least Squares

Three factorizations of one return matrix: a projection that rebuilds TAIEX from its sectors, a decomposition that measures rank and conditioning, and a triangular factor that whitens returns.

VOLUME II · Linear Algebra for Portfolios

BUILDS ON LM2 · LM4 · LM5 · LM7 · LM8 · LM9 · LM10 · LM11

LEADS TO LM16 · LM18 · LM23 · LM27 · LM30 · LM32

TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	**1** derive the normal equations from the orthogonality of the residual, and compute least-squares fits, projection matrices and R-squared
☐	**2** explain why TAIEX's daily returns are almost an exact combination of its sector indices' returns, and why the fitted coefficients are not sector weights
☐	**3** derive the singular value decomposition from the eigen-decomposition of $M^\top M$, and relate it to principal components and condition numbers
☐	**4** compute the minimum-norm solution of a rank-deficient least-squares problem with the pseudoinverse, and state a numerical rank with its tolerance and scale
☐	**5** compute a Cholesky factor by recursion, and interpret its entries as the slopes and residual standard deviations of sequential regressions
☐	**6** whiten and colour returns with a Cholesky factor, and explain why whitened returns have a mean squared length equal to the number of assets
☐	**7** apply `lstsq`, `pinv`, `svd`, `matrix_rank`, `cond` and `cholesky` in NumPy, and label each result as descriptive, in-sample or a stability check

LEARNING MODULE OVERVIEW

- Least squares projects a response on the column space of the design. The residual is orthogonal to every column, which gives the normal equations $X^\top X\hat{\boldsymbol{\beta}} = X^\top \mathbf{y}$; with an intercept, R-squared is $\cos^2\theta$.
- TAIEX's last 250 daily returns, fitted on the returns of 28 sector indices, give R-squared 0.99912 and a Semiconductor coefficient of 0.542. The fit rests on an accounting identity with drifting weights: the coefficients are the constant mix that best reproduces the window, not the sector weights.
- Fitted year by year, Semiconductor's coefficient rose from 0.189 in 2009 to 0.554 in 2026. Every year's R-squared is at least 0.99800, while one constant mix for the full sample reaches only 0.98333.
- Every matrix factors as $USV^\top$. For data, $\tilde{X}/\sqrt{T} = USV^\top$ gives $\hat{\Sigma} = VS^2V^\top$, so $\lambda_k = s_k^2$, and forming $\tilde{X}^\top\tilde{X}$ squares the condition number: 10.40 becomes 108.09 for the 28 sectors.
- December 2008 has 22 returns for 29 coefficients: rank 22, a null space of dimension 7 and an exact fit of any response. The pseudoinverse picks the minimum-norm fit, and the fit stays exact without Semiconductor.
- Numerical rank depends on the tolerance and the scale. Electronic and its eight sub-industries have rank 9 at $\delta = 0.01$ over the full sample and 8 in 2016, while `matrix_rank` reports 9 in every year.
- A positive definite $\hat{\Sigma}$ factors uniquely as $LL^\top$, a chain of regressions. Whitened returns $\mathbf{z}_t = L^{-1}(\mathbf{R}_t - \bar{\mathbf{R}})$ have identity covariance and a mean squared length of exactly N.
- For five sectors, 7 April 2025 has $\|\mathbf{z}_t\|^2 = 89.30$, the largest of 4,382 days, in either asset order; the 10% daily limit capped that day's returns, so even this figure understates the shock.

1 INTRODUCTION

Volume II has built vectors (LM8), linear systems (LM9), covariance matrices (LM10) and eigenvectors (LM11). This module closes the volume with three factorizations that software uses every day. Least squares splits a response into a fitted part and a residual orthogonal to it. The singular value decomposition (SVD) measures rank and conditioning on the data matrix itself. The Cholesky factor of a covariance matrix turns correlated returns into uncorrelated ones.

MOTIVATING CASE **Can you rebuild TAIEX from its sectors?**

A desk analyst wants to know how much of TAIEX moves with semiconductors, without a file of index weights. FinMind serves daily closes of 30 TWSE sector indices, and 28 of them contain no other sector index. She regresses TAIEX's last 250 daily returns, 25 September 2025 to 8 October 2026, on the returns of these 28 sectors, with an intercept.

The fit reproduces TAIEX almost perfectly: R-squared (Section 2) is 0.99912. The coefficient on Semiconductor is 0.542, on ElectronicPartsComponents 0.098, on FinancialInsurance 0.084 and on ComputerPeripheralEquipment 0.054. The 28 coefficients sum to 0.990 (Exhibit 1).

A colleague runs the same regression on the 22 daily returns of December 2008. That fit is exact, with R-squared 1, and it stays exact when the Semiconductor column is removed. Which fit tells more, and why is the perfect one worthless?

Exhibit 1: TAIEX regressed on 28 sector indices in three samples

	Full sample	Last 250 returns	December 2008
Daily returns T	4,382	250	22
Columns of the design $(\mathbf{1}\ X)$	29	29	29
Rank of the design	29	29	22
R-squared	0.98333	0.99912	1 (exact)
Sum of the 28 slopes	1.009	0.990	0.968
Semiconductor	0.337	0.542	0.145
ElectronicPartsComponents	0.081	0.098	0.065
FinancialInsurance	0.123	0.084	0.165
ComputerPeripheralEquipment	0.067	0.054	0.060
OtherElectronic	0.060	0.047	0.055
CommunicationsInternet	0.075	0.038	0.081

Note: least squares of TAIEX's daily simple returns on the 28 sector returns of U28, with an intercept. The rows show the six largest last-250 coefficients. Samples: 2 December 2008 → 8 October 2026, 25 September 2025 → 8 October 2026 and 2–31 December 2008. December 2008 has more coefficients than returns; its column shows the minimum-norm fit of Section 5, one of infinitely many exact fits. Source: FinMind `TaiwanStockPrice`, retrieved 11 October 2026, with 524 copied closes on 38 dates in this sample repaired from the 13:30:00 values of `TaiwanStockEvery5SecondsIndex` (`data/vol2_series.py`). Computations: `data/lm12_numbers.py` and `LM12_lab.ipynb`.

Section 2 treats least squares as a projection, and Section 3 applies it to the case. Section 4 derives the SVD. Section 5 handles rank deficiency and resolves the case, and Section 6 builds the Cholesky factor and whitens real returns.

NOTE **Notation for this module**

Returns are daily simple returns of price indices. The sample is 2 December 2008 → 8 October 2026, $T = 4{,}382$ returns (a count), unless a sub-sample is named. U28 is the 28 leaf indices among the 30 TWSE sector indices that FinMind serves daily. A leaf index contains no other sector index, so U28 excludes the aggregates Electronic and ChemicalBiotechnologyMedicalCare. S3 is Semiconductor, FinancialInsurance and Food; S5 adds Plastics and ShippingTransportation, in that order.

Local symbols: y_t TAIEX's return, $\bar{y}$ its mean, $\tilde{\mathbf{y}} = \mathbf{y} - \bar{y}\mathbf{1}$, $(\mathbf{1}\ X)$ the design with an intercept, $\hat{\alpha}$ the intercept, $\mathbf{x}$ a single column, $\mathrm{MV}_{i,t}$ the market value of index member i; m_{ij} and k_{ij} the entries of matrices M and K; $\mathbf{u}_k$ and $\mathbf{v}_k$ the columns of U and V, k the index of s_k and λ_k, s_k the singular values (not the standard deviations s_i of LM8 and LM10), M^{+} the pseudoinverse, $\kappa(M) = s_1/s_N$, δ a rank tolerance, $\operatorname{col} M$ and $\operatorname{null} M$ the column space and null space, $\mathbf{x}_h$ a vector of $\operatorname{null} M$; L the lower-triangular Cholesky factor, $\hat{\Sigma} = LL^{\top}$ (this module has no leverage), l_{ij} its entries, p their summation index (not m, which names entries of M), d_j the pivots of LM9 and LM10, $\hat{\sigma}_{ij}$ the entries of $\hat{\Sigma}$ and $\mathbf{z}_t = L^{-1}(\mathbf{R}_t - \bar{\mathbf{R}})$ the whitened returns, with entries $z_{j,t}$; σ_1^2, σ_{12} and σ_2^2 the entries of LM7's annualized covariance matrix of 0050 and 00679B (Problems 9 and 10). Each $\mathbf{v}_k$ is signed so that its entry largest in absolute value is positive (the first of tied entries), and $\mathbf{u}_k = M\mathbf{v}_k/s_k$. Moments divide by T, as in LM10.

NOTE **Research hygiene**

The editor fixed U28, S3 and S5 before this module was written. Every fit here is **in-sample**: an R-squared near 1 comes from an accounting identity, not from a forecast. The yearly fits are a **stability check**, not an out-of-sample test. The SVD, L and $\mathbf{z}_t$ use the full-sample $\hat{\Sigma}$, so they are **descriptive**: the most unusual day is judged with data from later years.

2 LEAST SQUARES AS A PROJECTION

- ☐ derive the normal equations from the orthogonality of the residual, and compute least-squares fits, projection matrices and R-squared

Least squares is geometry before it is algebra, so start with one column in the plane.

EXAMPLE 1

The closest multiple of one vector

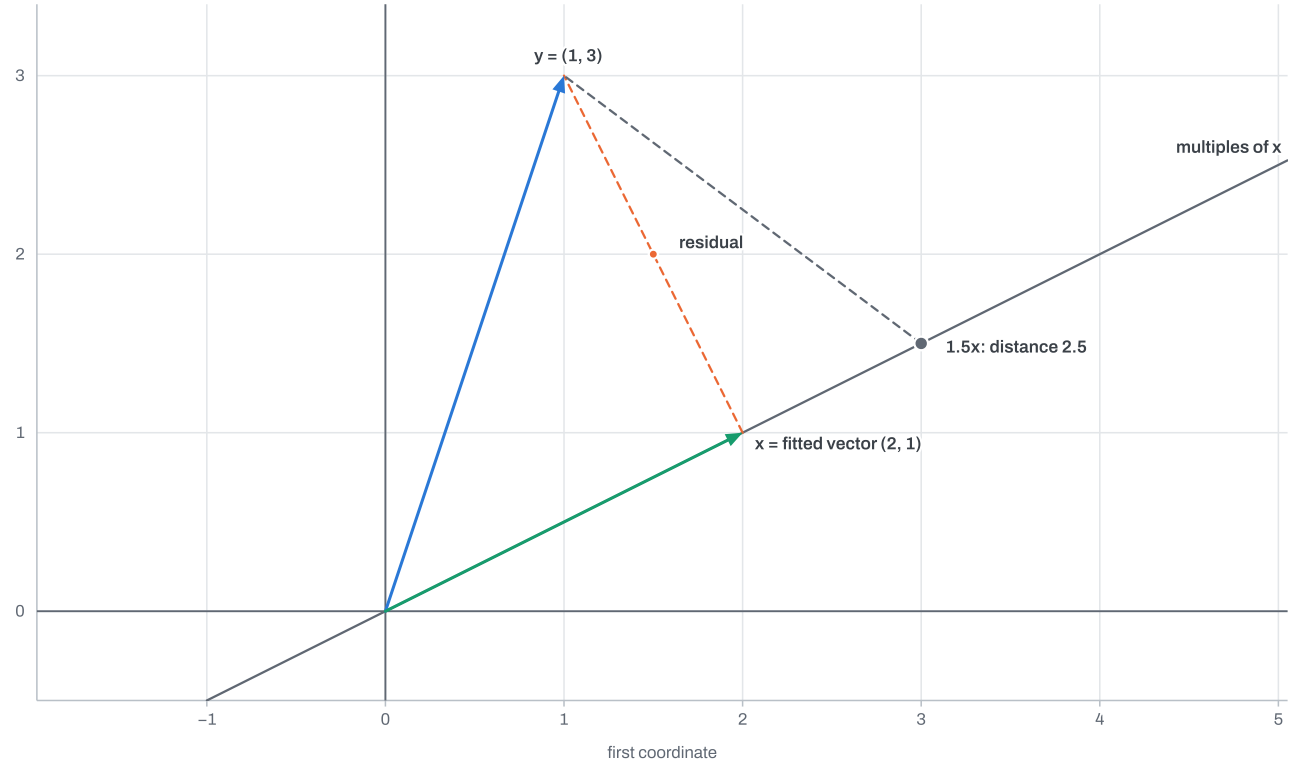
Find the multiple $\beta\mathbf{x}$ of $\mathbf{x} = (2, 1)^{\top}$ that is closest to $\mathbf{y} = (1, 3)^{\top}$, its residual and the distance.

SOLUTION

The multiples of $\mathbf{x}$ form a line through the origin (Exhibit 2). The point of the line closest to $\mathbf{y}$ is the foot of the perpendicular, so the residual must be orthogonal to $\mathbf{x}$: $\mathbf{x}^\top(\mathbf{y} - \beta\mathbf{x}) = 0$. Hence $\hat{\beta} = \mathbf{x}^\top\mathbf{y}/\mathbf{x}^\top\mathbf{x} = 5/5 = 1$.

The fitted vector is $\hat{\mathbf{y}} = (2, 1)^\top$, and the residual is $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{y}} = (-1, 2)^\top$, with $\mathbf{x}^\top\hat{\boldsymbol{\varepsilon}} = -2 + 2 = 0$. The distance is $\|\hat{\boldsymbol{\varepsilon}}\| = \sqrt{5} = 2.236$. Every other multiple is farther away: $1.5\mathbf{x}$ lies at distance 2.5. LM8 found the same closest multiple in its proof of (8.6), with the roles of the two vectors exchanged.

Exhibit 2: Least squares as a projection



Note: the line is the column space of the one-column matrix $\mathbf{x}$. The fitted vector $\hat{\mathbf{y}} = 1 \cdot \mathbf{x}$ is the foot of the perpendicular from $\mathbf{y}$. The residual (dashed) is orthogonal to the line, and the candidate $1.5\mathbf{x}$ is farther from $\mathbf{y}$.

The normal equations

Now let X be any $T \times N$ matrix with columns $\mathbf{x}_1, \ldots, \mathbf{x}_N$. Its **column space** $\operatorname{col} X$ is the set of all combinations $X\boldsymbol{\beta} = \beta_1\mathbf{x}_1 + \cdots + \beta_N\mathbf{x}_N$ (LM9). Least squares finds the point of $\operatorname{col} X$ closest to a response $\mathbf{y}$ (Strang 2016, §4.2). It minimizes $\|\mathbf{y} - X\boldsymbol{\beta}\|^2$, the sum of the squared misses over the T rows.

THEOREM **Least squares**

Let X be a $T \times N$ matrix and $\mathbf{y}$ a vector in $\mathbb{R}^T$. A vector $\hat{\boldsymbol{\beta}}$ minimizes $\|\mathbf{y} - X\boldsymbol{\beta}\|^2$ if and only if its residual $\hat{\boldsymbol{\varepsilon}} := \mathbf{y} - X\hat{\boldsymbol{\beta}}$ is orthogonal to every column of X:

$$X^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}, \quad \text{that is,} \quad X^\top X \hat{\boldsymbol{\beta}} = X^\top \mathbf{y}. \tag{12.1}$$

When X has full column rank, the minimizer is unique: $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$.

Proof. Suppose that $X^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$, and take any $\boldsymbol{\beta}$. Then $\mathbf{y} - X\boldsymbol{\beta} = \hat{\boldsymbol{\varepsilon}} + X(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ is a sum of two orthogonal vectors, and Pythagoras gives

$$\|\mathbf{y} - X\boldsymbol{\beta}\|^2 = \|\hat{\boldsymbol{\varepsilon}}\|^2 + \|X(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 \geq \|\hat{\boldsymbol{\varepsilon}}\|^2.$$

Conversely, expand the squared length and differentiate with the rules of LM10:

$$\begin{aligned} \|\mathbf{y} - X\boldsymbol{\beta}\|^2 &= \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top X^\top \mathbf{y} + \boldsymbol{\beta}^\top X^\top X \boldsymbol{\beta}, \\ \nabla_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|^2 &= 2X^\top X\boldsymbol{\beta} - 2X^\top \mathbf{y}. \end{aligned}$$

The gradient vanishes at any minimizer (LM7), which is (12.1). With full column rank, $X^\top X$ is positive definite (LM10), hence invertible. ■

The theorem names the **fitted values** $\hat{\mathbf{y}} := X\hat{\boldsymbol{\beta}}$, the point of $\operatorname{col} X$ closest to $\mathbf{y}$. Equations (12.1) are the **normal equations**, "normal" meaning perpendicular. Section 5 shows that they have a solution even without full column rank.

The projection matrix

With full column rank, $\hat{\mathbf{y}} = X(X^\top X)^{-1} X^\top \mathbf{y}$, so one matrix sends every response to its fitted values:

$$\begin{aligned} \Pi &:= X(X^\top X)^{-1} X^\top, \\ \Pi^\top &= \Pi = \Pi^2, \qquad \operatorname{tr} \Pi = N. \end{aligned} \tag{12.2}$$

The **projection matrix** Π is symmetric, by the rule $(MK)^\top = K^\top M^\top$ of LM8 and because the inverse of the symmetric matrix $X^\top X$ is symmetric. It is idempotent, $\Pi^2 = \Pi$, because the middle factor $X^\top X$ of Π^2 cancels one inverse: projecting twice changes nothing. Its trace counts the columns. For two matrices whose products KM and MK are both square, the trace is cyclic, $\operatorname{tr}(KM) = \sum_i \sum_j k_{ij} m_{ji} = \operatorname{tr}(MK)$. Here k_{ij} and m_{ij} are the entries of K and M. Hence

$$\operatorname{tr} \Pi = \operatorname{tr}\left((X^\top X)^{-1} X^\top X\right) = \operatorname{tr} I_N = N.$$

The residual is $\hat{\boldsymbol{\varepsilon}} = (I - \Pi)\mathbf{y}$, and $I - \Pi$ projects on the directions orthogonal to every column.

The intercept and R-squared

A fit with an intercept uses the design $(\mathbf{1}\ X)$, whose first column is $\mathbf{1}$. Its first normal equation, $\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = 0$, says that the residuals sum to zero. Averaging $\mathbf{y} = \hat{\alpha}\mathbf{1} + X\hat{\boldsymbol{\beta}} + \hat{\boldsymbol{\varepsilon}}$ over the rows therefore gives $\bar{y} = \hat{\alpha} + \bar{\mathbf{R}}^\top \hat{\boldsymbol{\beta}}$, and subtracting leaves $\tilde{\mathbf{y}} = \tilde{X}\hat{\boldsymbol{\beta}} + \hat{\boldsymbol{\varepsilon}}$. The other normal equations, $X^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$, combine with the first into $\tilde{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$. Substituting $\hat{\boldsymbol{\varepsilon}} = \tilde{\mathbf{y}} - \tilde{X}\hat{\boldsymbol{\beta}}$ and dividing by T:

$$\hat{\Sigma}\hat{\boldsymbol{\beta}} = \tfrac{1}{T}\tilde{X}^\top \tilde{\mathbf{y}}, \qquad \hat{\alpha} = \bar{y} - \bar{\mathbf{R}}^\top \hat{\boldsymbol{\beta}}. \tag{12.3}$$

The slopes depend only on covariances, and the intercept on the means and the slopes.

Centring splits $\tilde{\mathbf{y}}$ into two orthogonal pieces, the centred fit $\tilde{X}\hat{\boldsymbol{\beta}} = \hat{\mathbf{y}} - \bar{y}\mathbf{1}$ and the residual. **R-squared** is the share of the centred response's squared length that the fit reproduces (LM23 treats it as a statistic). By Pythagoras,

$$\begin{aligned} \text{R-squared} &:= 1 - \frac{\|\hat{\boldsymbol{\varepsilon}}\|^2}{\|\tilde{\mathbf{y}}\|^2} \\ &= \frac{\|\hat{\mathbf{y}} - \bar{y}\mathbf{1}\|^2}{\|\tilde{\mathbf{y}}\|^2} = \cos^2\theta, \end{aligned} \tag{12.4}$$

where θ is the angle between $\tilde{\mathbf{y}}$ and the centred fit (LM8). With one column, $\cos\theta$ is the absolute value of the sample correlation of $\mathbf{x}$ and $\mathbf{y}$, so R-squared is the squared correlation.

EXAMPLE 2

A line through four points

Fit $\mathbf{y} = \hat{\alpha}\mathbf{1} + \hat{\beta}\mathbf{x} + \hat{\boldsymbol{\varepsilon}}$ to $\mathbf{x} = (0, 1, 2, 3)^\top$ and $\mathbf{y} = (1, 3, 2, 6)^\top$. Check (12.1), and compute R-squared in two ways.

SOLUTION

The means are 1.5 and 3, so $\tilde{\mathbf{x}} = (-1.5, -0.5, 0.5, 1.5)^\top$ and $\tilde{\mathbf{y}} = (-2, 0, -1, 3)^\top$. By (12.3), $\hat{\beta} = \tilde{\mathbf{x}}^\top\tilde{\mathbf{y}}/\tilde{\mathbf{x}}^\top\tilde{\mathbf{x}} = 7/5 = 1.4$ and $\hat{\alpha} = 3 - 1.4 \times 1.5 = 0.9$.

The fitted values are $(0.9, 2.3, 3.7, 5.1)^\top$ and the residuals $(0.1, 0.7, -1.7, 0.9)^\top$. The residuals sum to zero, and $\mathbf{x}^\top\hat{\boldsymbol{\varepsilon}} = 0 + 0.7 - 3.4 + 2.7 = 0$, as (12.1) requires.

The squared lengths are $\|\hat{\boldsymbol{\varepsilon}}\|^2 = 4.2$ and $\|\tilde{\mathbf{y}}\|^2 = 14$, so R-squared is $1 - 4.2/14 = 0.7$. As a squared cosine it is $(\tilde{\mathbf{x}}^\top\tilde{\mathbf{y}})^2/(\|\tilde{\mathbf{x}}\|^2\|\tilde{\mathbf{y}}\|^2) = 49/70 = 0.7$.

PITFALL **Without an intercept, R-squared is another number**

Formula (12.4) needs $\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = 0$, which only an intercept guarantees. Fit Example 2's data through the origin instead: $\hat{\beta} = 25/14 = 1.786$ and $\|\hat{\boldsymbol{\varepsilon}}\|^2 = 5.357$, a worse fit than 4.2. Yet the uncentred share $1 - \|\hat{\boldsymbol{\varepsilon}}\|^2/\|\mathbf{y}\|^2 = 1 - 5.357/50 = 0.893$ exceeds 0.7. The centred formula gives 0.617, and no angle stands behind it. Compare fits by their residuals, and quote R-squared only for fits with an intercept.

KNOWLEDGE CHECK 1

In Example 1 the residual is orthogonal to the fitted vector. Why does the theorem demand orthogonality to every column of X, not only to $\hat{\mathbf{y}}$?

ANSWER

Orthogonality to the fitted vector alone also holds at bad points. At $\beta = 0$ the fit is $\mathbf{0}$, and every residual is orthogonal to $\mathbf{0}$, yet the distance $\|\mathbf{y}\| = \sqrt{10} = 3.162$ exceeds the minimum $\sqrt{5}$. If a column has $\mathbf{x}_j^\top \hat{\boldsymbol{\varepsilon}} \neq 0$, the gradient is not zero, and a small step along that column shortens the residual.

3 READING TAIEX FROM ITS SECTORS

- [ ] explain why TAIEX's daily returns are almost an exact combination of its sector indices' returns, and why the fitted coefficients are not sector weights

Before reading the case's coefficients, consider why such a fit can be nearly exact.

The index identity

TWSE computes TAIEX as the total market value of its member stocks divided by a base value, times the base level (TWSE IndexS02). It computes the sector indices the same way (TWSE IndexS03). Suppose nothing but prices changes between two closes, so that the market value $\mathrm{MV}_{i,t}$ of member i grows by its return, $\mathrm{MV}_{i,t} = \mathrm{MV}_{i,t-1}(1 + R_{i,t})$. With the same base value on both days,

$$1 + y_t = \frac{\sum_i \mathrm{MV}_{i,t}}{\sum_i \mathrm{MV}_{i,t-1}} = \sum_i w_{i,t-1}(1 + R_{i,t}),$$
$$w_{i,t-1} := \frac{\mathrm{MV}_{i,t-1}}{\sum_j \mathrm{MV}_{j,t-1}}.$$

So TAIEX's return is a weighted average of its members' returns, $\mathbf{w}_{t-1}^\top \mathbf{R}_t$ as in LM8, with weights set at the previous close. The same identity holds inside each sector index. Suppose every member of TAIEX belongs to exactly one of the 28 sectors. Grouping the members by sector then turns TAIEX's return into an average of sector returns, weighted by the sectors' market values at the previous close.

Two things break the identity: members or share counts that change, which alter the base value, and members outside the 28 sectors. TWSE added four sector indices with base date 30 June 2023 (綠能環保類, 數位雲端類, 運動休閒類 and 居家生活類; TWSE IndexS03). FinMind serves no daily series for them, so from July 2023, which includes the whole last-250 window, some TAIEX members lie outside U28.

A third thing, weights that drift every day with relative prices, leaves the identity intact. The drift keeps any constant mix from reproducing the identity. The coefficients of a fit are therefore the constant weights that best reproduce TAIEX's daily moves in the window, never "the sector weights".

EXAMPLE 3

Resolving the case: the last 250 returns

Fit TAIEX's last 250 daily returns on the 28 sector returns with an intercept, and read the coefficients.

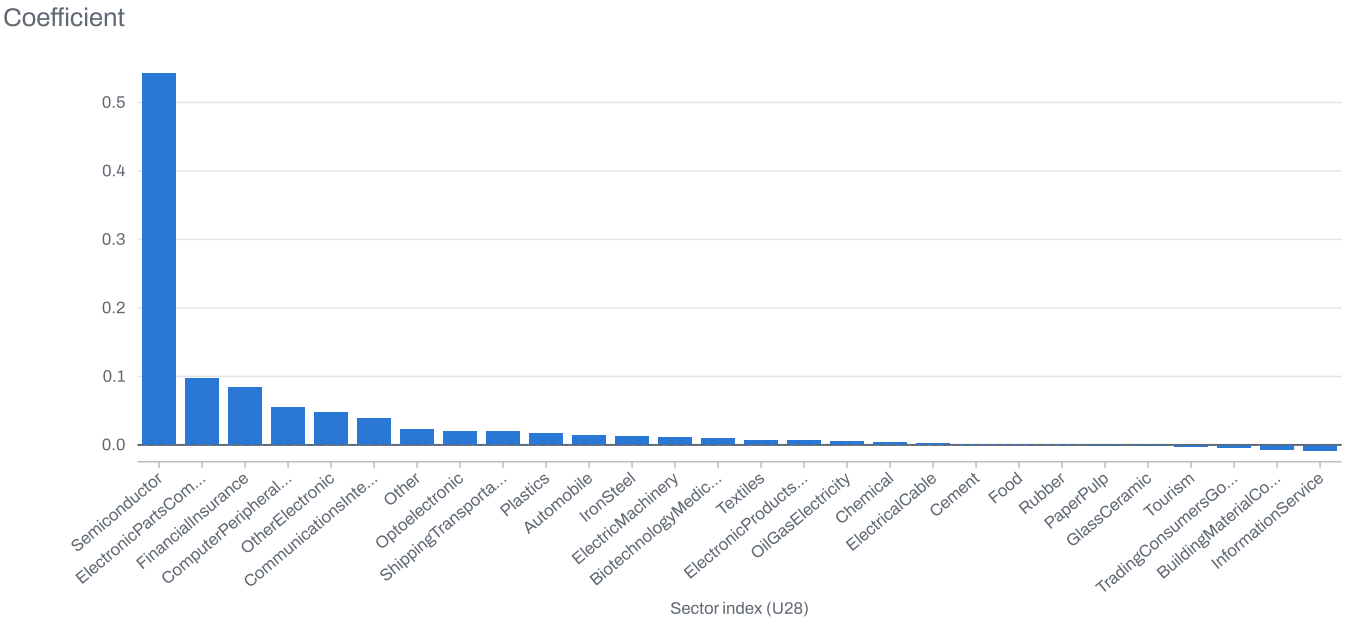
SOLUTION

The design $(\mathbf{1}\ X)$ is 250×29 with full column rank, so (12.1) has one solution. R-squared is 0.99912: the fit reproduces 99.912% of the centred response's squared length. Semiconductor's coefficient is 0.542, followed by ElectronicPartsComponents 0.098, FinancialInsurance 0.084, ComputerPeripheralEquipment 0.054, OtherElectronic 0.047 and CommunicationsInternet 0.038 (Exhibit 3).

The 28 coefficients sum to 0.990, and the intercept is −0.013% a day; this module does not explain either number. Seven coefficients are negative, the smallest −0.0081 (InformationService). A market value cannot be negative, so these coefficients are not weights: they are fitting noise among correlated columns.

Semiconductor matters most: without its column, R-squared falls to 0.856. So the analyst's answer describes the window: a constant mix with 0.542 in Semiconductor reproduced TAIEX's daily moves best.

Exhibit 3: Coefficients of the 28 sectors, last 250 returns



Note: least-squares coefficients of TAIEX's daily returns on the 28 sector returns with an intercept, 25 September 2025 → 8 October 2026, sorted; R-squared 0.99912. Seven coefficients are negative.

EXAMPLE 4

One fit per year

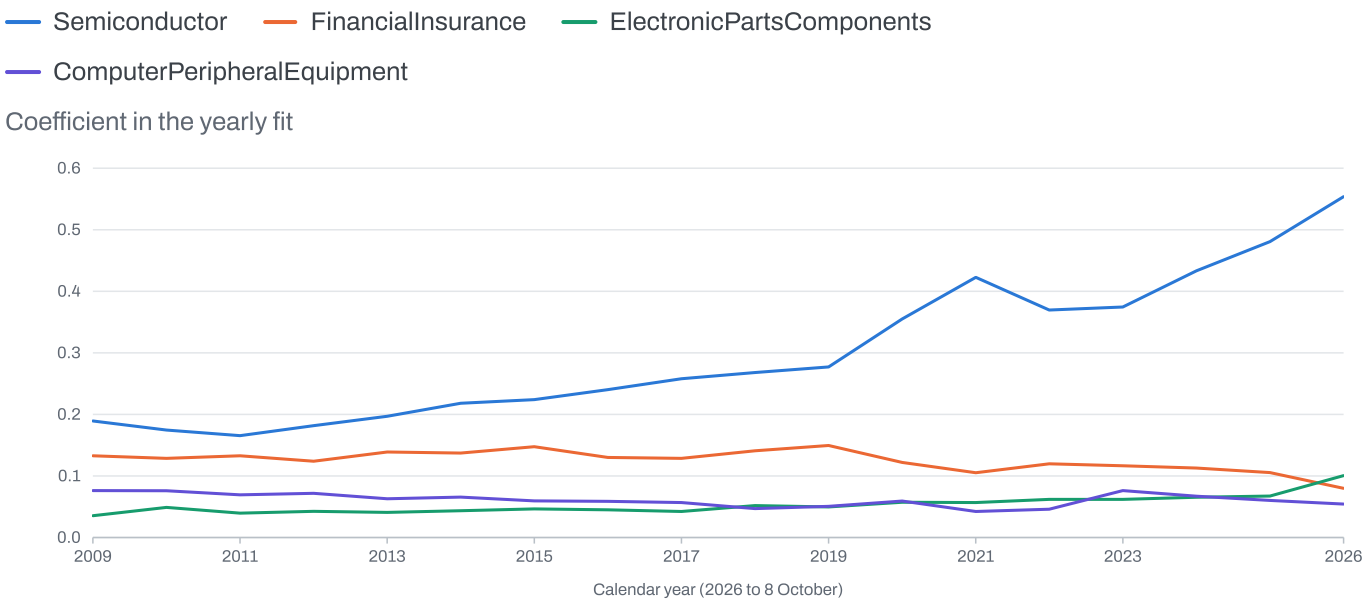
Repeat the fit on each calendar year from 2009 to 2026 (to 8 October) and on the full sample. How stable are the coefficients, and how good is each fit?

SOLUTION

Semiconductor's coefficient rises from 0.189 in 2009 to 0.554 in 2026, and FinancialInsurance's falls from 0.133 to 0.080 (Exhibit 4). Every year's R-squared is at least 0.99800, the value of 2021. The full sample, one constant mix for almost 18 years, reaches only 0.98333. A constant mix cannot follow coefficients that drift this much, so the full-sample fit is worse than every yearly fit.

The comparison is a stability check. It shows how the best-reproducing mix changed; it does not show how well any mix predicts.

Exhibit 4: Four coefficients by calendar year

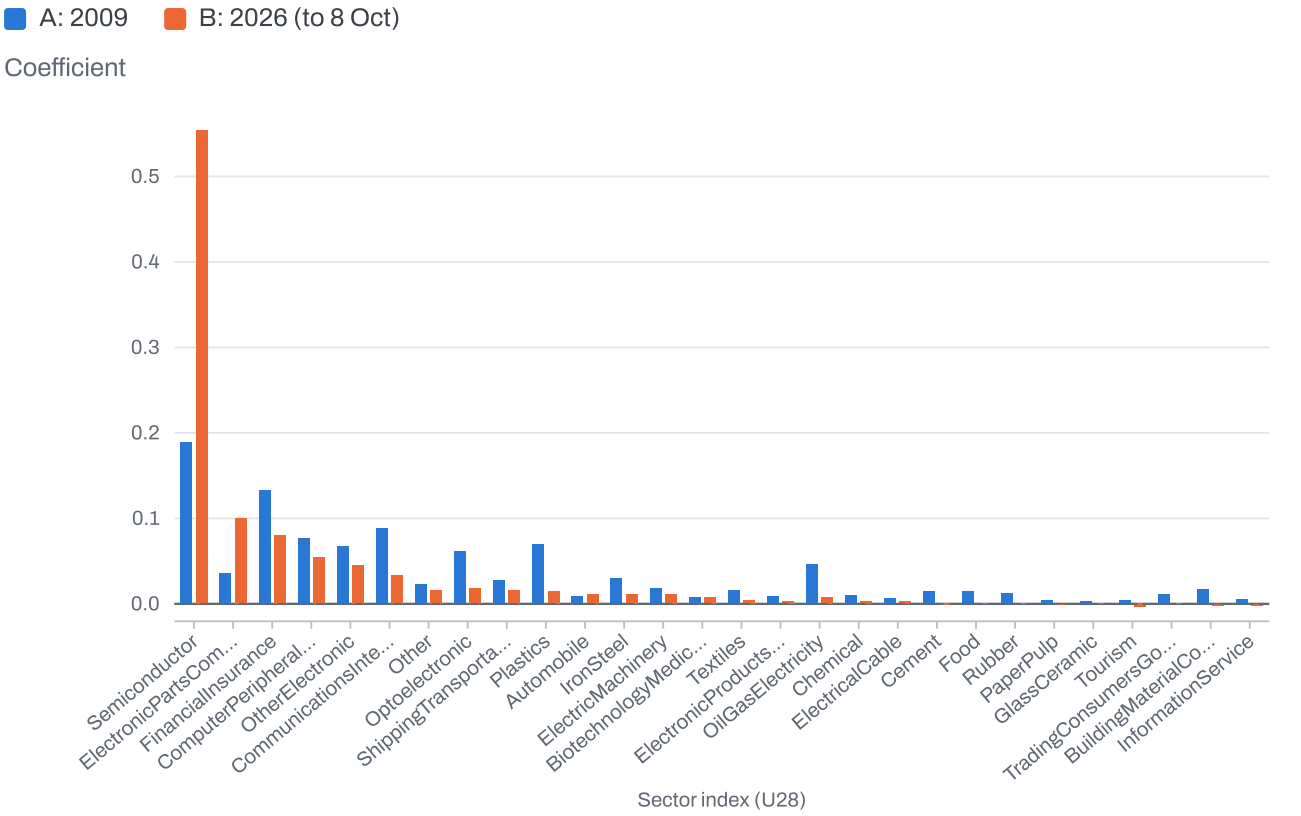


Note: coefficients of one least-squares fit per calendar year of TAIEX's daily returns on the 28 sector returns, with an intercept. 2026 holds the 185 returns to 8 October.

The widget compares the 28 coefficients of any two samples: a calendar year, the last 250 returns or the full sample.

Rebuilding TAIEX from its sectors: the 28 coefficients in two samples

INTERACTIVE



A: returns · R-squared · sum of slopes
251 · 0.99959 · 1.008

B: returns · R-squared · sum of slopes
185 · 0.99952 · 0.987

Semiconductor, A → B
0.189 → 0.554

Largest change, B – A
Semiconductor +0.365

Negative coefficients, A · B
0 · 5

Each sample has its own least-squares fit of TAIEX's daily returns on the 28 sector returns, with an intercept. The sectors keep the order of the last-250 fit. Every fit is in-sample, and comparing two years is a stability check, not a forecast; 2026 holds the returns to 8 October.

導讀 **最小平方法：投影，不是權重**

最小平方法（least squares）把 $\mathbf{y}$ 投影（projection）到設計矩陣（design matrix）的行空間（column space）上：殘差（residual）必須和每一行都正交（orthogonal），這就是正規方程（normal equations）。加上截距（intercept）以後，判定係數（R-squared）等於中心化（centring）後的 $\mathbf{y}$ 與中心化配適值（fitted value）夾角（angle）餘弦（cosine）的平方。加權股價指數的報酬本來就是成分股報酬的市值加權平均，所以對 28 個類股指數迴歸（regression），判定係數幾乎是 1；這是會計恆等式（accounting identity），不是預測能力。係數是「這段期間最能重現大盤每日漲跌的固定權重」，不是類股市值權重：真正的權重每天隨股價漂移，而且有七個係數是負的。逐年重做時半導體的係數從 0.189 升到 0.554，這是穩定性檢查（stability check），不是樣本外檢定（out-of-sample test）。

KNOWLEDGE CHECK 2

Why can a least-squares coefficient be slightly negative when every sector's market-value weight is positive?

ANSWER

Least squares minimizes the squared misses and imposes no sign. The true weights drift, and correlated columns such as the electronics sub-industries partly substitute for one another. A small negative coefficient can therefore lower the sum of squared residuals. In the last 250 returns, seven coefficients are negative, the smallest −0.0081 (InformationService).

4 THE SINGULAR VALUE DECOMPOSITION

- ☐ derive the singular value decomposition from the eigen-decomposition of $M^{\top}M$, and relate it to principal components and condition numbers

LM11 decomposed the symmetric matrix $\hat{\Sigma}$. The SVD decomposes a rectangular data matrix directly.

From a circle to an ellipse

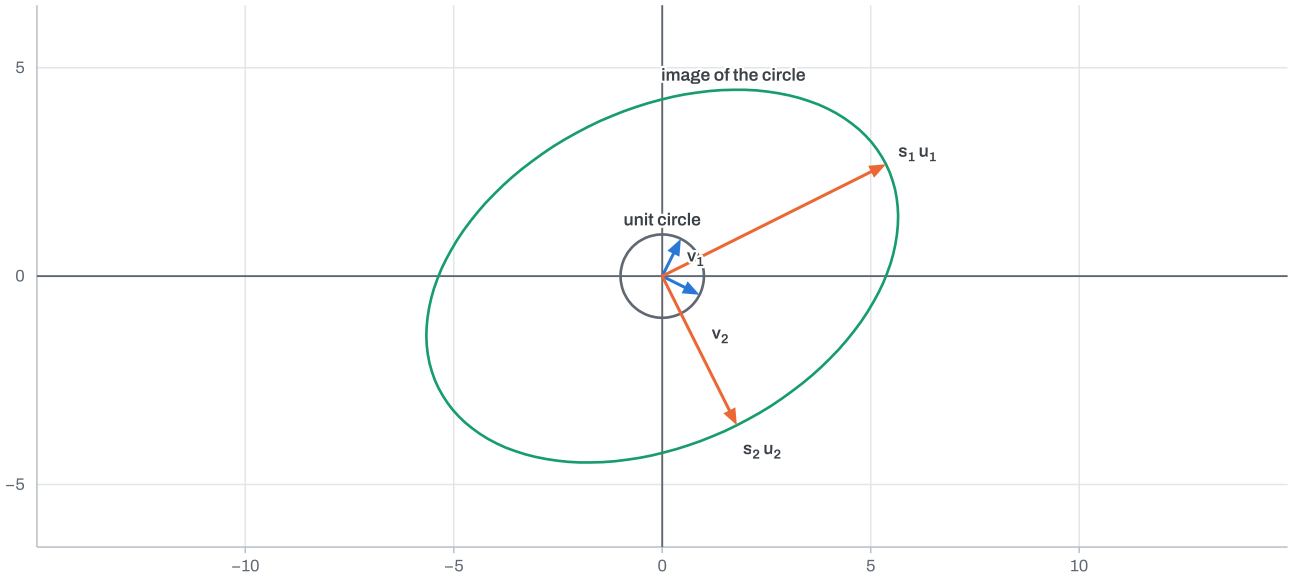
An invertible 2×2 matrix maps the unit circle to an ellipse. For

$$M = \begin{pmatrix} 4 & 4 \\ -2 & 4 \end{pmatrix},$$

the ellipse has semi-axes of lengths 6 and 4 (Exhibit 5). Two perpendicular unit vectors, $\mathbf{v}_1$ and $\mathbf{v}_2$, map onto the axes: $M\mathbf{v}_k = s_k\mathbf{u}_k$, where $\mathbf{u}_1$ and $\mathbf{u}_2$ are perpendicular unit vectors and $s_1 = 6$, $s_2 = 4$. So $M =$

$USV^\top$ acts in three steps: $V^\top$ carries each $\mathbf{v}_k$ to a coordinate axis, S stretches the axes by s_k, and U carries them to the $\mathbf{u}_k$.

Exhibit 5: An invertible 2×2 matrix maps the unit circle to an ellipse



Note: M of Example 5. The unit circle (grey) maps to the ellipse with semi-axes $s_1 = 6$ and $s_2 = 4$. The right singular vectors $\mathbf{v}_1, \mathbf{v}_2$ map to $s_1\mathbf{u}_1$ and $s_2\mathbf{u}_2$, the axes of the ellipse.

The decomposition

Let M be a real $T \times N$ matrix. The matrix $M^\top M$ is symmetric and positive semidefinite, since $\boldsymbol{\beta}^\top M^\top M \boldsymbol{\beta} = \|M\boldsymbol{\beta}\|^2 \geq 0$ for every $\boldsymbol{\beta}$. By the spectral theorem of LM11, it has orthonormal eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_N$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_N \geq 0$. Set $s_k := \sqrt{\lambda_k}$ and, when $s_k > 0$, $\mathbf{u}_k := M\mathbf{v}_k / s_k$. These vectors are orthonormal:

$$\mathbf{u}_j^\top \mathbf{u}_k = \frac{\mathbf{v}_j^\top M^\top M \mathbf{v}_k}{s_j s_k} = \frac{\lambda_k \, \mathbf{v}_j^\top \mathbf{v}_k}{s_j s_k} = \begin{cases} 1, & j = k, \\ 0, & j \neq k. \end{cases}$$

When $s_k = 0$, $\|M\mathbf{v}_k\|^2 = \lambda_k = 0$, so $M\mathbf{v}_k = \mathbf{0}$. Because the $\mathbf{v}_k$ form an orthonormal basis, $M = \sum_k M\mathbf{v}_k\mathbf{v}_k^\top$, and only the terms with $s_k > 0$ survive.

THEOREM **Singular value decomposition**

Let M be a real $T \times N$ matrix, with singular values $s_1 \geq s_2 \geq \cdots \geq 0$ (the square roots of the eigenvalues of $M^\top M$) and orthonormal eigenvectors $\mathbf{v}_k$ of $M^\top M$. Then

$$M = \sum_{s_k > 0} s_k \, \mathbf{u}_k \mathbf{v}_k^\top, \qquad \mathbf{u}_k = M\mathbf{v}_k / s_k,$$

where the $\mathbf{u}_k$ are orthonormal in $\mathbb{R}^T$ and the number of positive s_k is rank M. When rank $M = N$, stacking the pieces gives the **thin SVD** $M = USV^\top$: U is $T \times N$ with $U^\top U = I_N$, $S = \operatorname{diag}(s_1, \ldots, s_N)$ and V is orthogonal.

The number of positive s_k is the rank, because $M\boldsymbol{\beta} = \sum_{s_k>0} s_k (\mathbf{v}_k^\top \boldsymbol{\beta})\, \mathbf{u}_k$ ranges over the span of those $\mathbf{u}_k$ (Strang 2016, ch. 7).

EXAMPLE 5

The SVD of a 2×2 matrix

Compute the SVD of the matrix M above by hand.

SOLUTION

First,

$$M^\top M = \begin{pmatrix} 20 & 8 \\ 8 & 32 \end{pmatrix},$$

with trace 52 and determinant 576. By LM11's 2×2 formula, the eigenvalues are $26 \pm \sqrt{26^2 - 576} = 26 \pm 10$: $\lambda_1 = 36$ and $\lambda_2 = 16$, so $s_1 = 6$ and $s_2 = 4$.

For $\lambda_1 = 36$ the first row of $M^\top M - 36I$ is $(-16, 8)$, so $\mathbf{v}_1 = (1, 2)^\top / \sqrt{5}$. Orthogonality and the sign rule give $\mathbf{v}_2 = (2, -1)^\top / \sqrt{5}$. Then $\mathbf{u}_1 = M\mathbf{v}_1 / 6 = (2, 1)^\top / \sqrt{5}$ and $\mathbf{u}_2 = M\mathbf{v}_2 / 4 = (1, -2)^\top / \sqrt{5}$.

Check: $|\det M| = |16 + 8| = 24 = s_1 s_2$, because the orthogonal factors have determinants ± 1.

Principal components from the data matrix

Apply the SVD to the centred data matrix, scaled by $1/\sqrt{T}$:

$$\frac{\tilde{X}}{\sqrt{T}} = USV^\top \implies \hat{\Sigma} = \frac{\tilde{X}^\top \tilde{X}}{T} = V S^2 V^\top. \tag{12.5}$$

Compare LM11's $\hat{\Sigma} = Q\Lambda Q^\top$. The eigenvalues are the squared singular values, $\lambda_k = s_k^2$. When the eigenvalues are distinct, the right singular vectors are the principal directions, $\mathbf{v}_k = \mathbf{q}_k$ under the same

sign rule. The SVD also gives the score series: $\tilde{X}\mathbf{v}_k = \sqrt{T}\, s_k \mathbf{u}_k$. This route never forms $\tilde{X}^\top \tilde{X}$, which matters for accuracy.

Conditioning

For a matrix of full column rank, the **condition number** is $\kappa(M) := s_1/s_N$. On a symmetric positive definite matrix the singular values are the eigenvalues, so κ agrees with LM11's λ_1/λ_N. Because $\tilde{X}^\top \tilde{X} = T\, V S^2 V^\top$,

$$\kappa(\tilde{X}^\top \tilde{X}) = \kappa(\hat{\Sigma}) = \frac{s_1^2}{s_N^2} = \kappa(\tilde{X})^2. \tag{12.6}$$

The normal equations work with $\tilde{X}^\top \tilde{X}$, so they square the condition number. Floating-point arithmetic carries about 16 significant digits (LM2), so rounding changes a matrix by a relative amount of order 10^{-16}. By LM11's bound (11.9), such a change in a positive definite matrix moves the solution by a relative amount of at most about $\kappa \times 10^{-16}$. Hence, as a rule of thumb, a condition number κ can cost about $\log_{10} \kappa$ of the 16 digits.

`np.linalg.lstsq` works on the matrix it is given and returns that matrix's singular values (NumPy reference). A call on $\tilde{X}$ therefore never squares $\kappa(\tilde{X})$ by forming $\tilde{X}^\top \tilde{X}$. When the residual is small, the accuracy of its slopes depends on $\kappa(\tilde{X})$ rather than on $\kappa(\tilde{X})^2$.

EXAMPLE 6

Conditioning of the 28 sectors

Compute the extreme singular values of $\tilde{X}/\sqrt{T}$ for U28 on the full sample, and the condition numbers of $\tilde{X}$ and $\hat{\Sigma}$. Repeat for the last 250 returns.

SOLUTION

The largest singular value is $s_1 = 0.054245$ and the smallest is $s_{28} = 0.005218$, so $\kappa(\tilde{X}) = 10.40$. By (12.6), $\kappa(\hat{\Sigma}) = 108.09$ from the unrounded ratio, the figure LM11 finds from the eigenvalues. On the last 250 returns, $\kappa(\tilde{X}) = 17.04$ and $\kappa(\tilde{X}^\top \tilde{X}) = 290.2$.

Both are small, and Exhibit 6 shows that `lstsq` and the normal equations agree to better than 10^{-14}.

Exhibit 6: Condition numbers and the gap between two solvers

Centred columns, sample	Columns	$\kappa(\tilde{X})$	$\kappa(\tilde{X}^\top \tilde{X})$	Largest gap
S3, full sample	3	2.654	7.045	below 10^{-14}
S5, full sample	5	3.481	12.12	below 10^{-14}
U28, full sample	28	10.40	108.09	below 10^{-14}
U28, last 250 returns	28	17.04	290.2	below 10^{-14}
U28 + Electronic, full sample	29	48.33	2,336	below 10^{-12}
U28 + Electronic, 2016	29	642.0	412,167	10^{-12} to 10^{-11}

Note: the response is TAIEX's centred daily returns. The gap is the largest absolute difference between the slopes from `np.linalg.lstsq` on $\tilde{X}$ and from `np.linalg.solve` on the normal equations. Only its order matters, since it depends on the machine. The rule of thumb puts the loss at $\log_{10} \kappa(\tilde{X}^\top \tilde{X})$ digits: 2.0 for U28 and 5.6 for U28 + Electronic in 2016.

KNOWLEDGE CHECK 3

A 2×2 matrix M has singular values 5 and 0.5. What are its condition number and the absolute value of its determinant? What are they for $M^\top M$?

ANSWER

The condition number is $5/0.5 = 10$ and $|\det M| = 5 \times 0.5 = 2.5$. The matrix $M^\top M$ has eigenvalues 25 and 0.25, so its condition number is 100, the square of 10, and its determinant is 6.25.

5 RANK DEFICIENCY, NUMERICAL RANK AND THE PSEUDOINVERSE

- ☐ compute the minimum-norm solution of a rank-deficient least-squares problem with the pseudoinverse, and state a numerical rank with its tolerance and scale
- ☐ apply `lstsq`, `pinv`, `svd`, `matrix_rank`, `cond` and `cholesky` in NumPy, and label each result as descriptive, in-sample or a stability check

More coefficients than returns

December 2008 has 22 daily returns, and the design $(\mathbf{1}\ X)$ has 29 columns. Its rank is at most 22, and NumPy finds exactly 22, so by (9.7) of LM9 the null space has dimension $29 - 22 = 7$. The columns then

span all of $\mathbb{R}^{22}$: every response, TAIEX or not, is fitted exactly. R-squared is 1 by arithmetic, not by evidence.

The normal equations then have infinitely many solutions, because adding a vector of the null space changes no fitted value. They form LM9's particular solution plus null space, and software needs a rule to pick one.

The pseudoinverse

THEOREM **Minimum-norm least squares**

Let M be a real $T \times N$ matrix with the SVD of Section 4, and let $\mathbf{y}$ be in $\mathbb{R}^T$. Define the **pseudoinverse**

$$M^{+} := \sum_{s_k > 0} \frac{1}{s_k}\, \mathbf{v}_k \mathbf{u}_k^{\top}. \tag{12.7}$$

In matrix form $M^{+} = V S^{-1} U^{\top}$, where U, S and V keep only the positive singular values and their vectors. Then $M^{+}\mathbf{y}$ solves the normal equations $M^{\top}M\boldsymbol{\beta} = M^{\top}\mathbf{y}$, and every other solution has a larger norm.

The vector $M^{+}\mathbf{y}$ solves the normal equations. Transposing the theorem of Section 4 gives $M^{\top} = \sum_{s_k>0} s_k \mathbf{v}_k \mathbf{u}_k^{\top}$, and the orthonormal $\mathbf{u}_k$ then give $M^{\top}M = \sum_{s_k>0} s_k^2 \mathbf{v}_k \mathbf{v}_k^{\top}$. Since the $\mathbf{v}_k$ are orthonormal too,

$$M^{\top}M M^{+}\mathbf{y} = \sum_{s_k>0} s_k\, \mathbf{v}_k \mathbf{u}_k^{\top}\mathbf{y} = M^{\top}\mathbf{y}.$$

At full column rank, $M^{+} = (M^{\top}M)^{-1}M^{\top}$ gives the unique solution of Section 2.

DEEP DIVE **Why $M^{+}\mathbf{y}$ has the smallest norm** optional · click to expand

Two solutions of the normal equations differ by a vector $\mathbf{x}_h$ with $M^{\top}M\mathbf{x}_h = \mathbf{0}$, so $\|M\mathbf{x}_h\|^2 = \mathbf{x}_h^{\top}M^{\top}M\mathbf{x}_h = 0$. By Section 4, $M\mathbf{x}_h = \sum_{s_k>0} s_k(\mathbf{v}_k^{\top}\mathbf{x}_h)\,\mathbf{u}_k$ vanishes exactly when $\mathbf{x}_h$ is orthogonal to every $\mathbf{v}_k$ with $s_k > 0$. The vector $M^{+}\mathbf{y}$ is a combination of these $\mathbf{v}_k$, so it is orthogonal to $\mathbf{x}_h$, and Pythagoras gives

$$\|M^{+}\mathbf{y} + \mathbf{x}_h\|^2 = \|M^{+}\mathbf{y}\|^2 + \|\mathbf{x}_h\|^2.$$

Every other solution is longer, and $M^{+}\mathbf{y}$ is the only solution orthogonal to null M.

EXAMPLE 7

December 2008: two exact fits

Fit TAIEX's 22 daily returns of December 2008 on the 28 sectors with an intercept. Compare the minimum-norm solution with the minimum-norm fit after deleting Semiconductor's column.

SOLUTION

The design is 22×29 with rank 22, so every fit is exact. `np.linalg.lstsq` and `np.linalg.pinv` both return the minimum-norm solution (12.7). Its Semiconductor coefficient is 0.145, FinancialInsurance's is 0.165, and the norm of all 29 coefficients is 0.285 (Exhibit 7).

Deleting Semiconductor leaves a 22×28 design, still of rank 22, so the fit stays exact. Its minimum-norm solution moves FinancialInsurance to 0.218, with norm 0.401. The two exact fits disagree on every coefficient. As LM9's three-day fits showed, an exact fit carries no evidence about weights; this answers the case's second question.

Exhibit 7: December 2008: the minimum-norm fit with and without Semiconductor

	All 28 sectors	Without Semiconductor
Design	22 × 29	22 × 28
Rank	22	22
Dimension of the null space	7	6
Semiconductor	0.145	—
FinancialInsurance	0.165	0.218
ElectronicPartsComponents	0.065	0.172
CommunicationsInternet	0.081	0.068
Sum of the slopes	0.968	0.914
Norm of all coefficients	0.285	0.401
Largest absolute residual	below 10^{-13}	below 10^{-13}

Note: TAIEX's 22 daily returns of 2–31 December 2008 on the sector returns with an intercept; minimum-norm solutions (12.7) from `np.linalg.pinv`. Both R-squared values are 1 to machine precision.

Numerical rank

In exact arithmetic the rank is the number of non-zero singular values. Floating-point arithmetic keeps about 16 significant digits (LM2), so a singular value that is zero in exact arithmetic is computed as a number of order $10^{-16} s_1$. Software therefore counts the singular values above a tolerance: the **numerical rank** at tolerance δ is the number of $s_k > \delta s_1$. By default, `np.linalg.matrix_rank` uses $\delta = \max(T, N) \times 2.2 \times 10^{-16}$, and `np.linalg.pinv` treats every $s_k \le 10^{-15} s_1$ as zero (NumPy reference).

Such a tolerance answers a question about arithmetic, not about data. LM11 found Electronic nearly a combination of its eight sub-industries, a relation that a tolerance near 10^{-15} cannot see. That near-

dependence is why U28 excludes the aggregates: adding Electronic raises $\kappa(\tilde{X})$ from 10.40 to 48.33, and to 642.0 in 2016 (Exhibit 6).

EXAMPLE 8

Electronic and its eight sub-industries

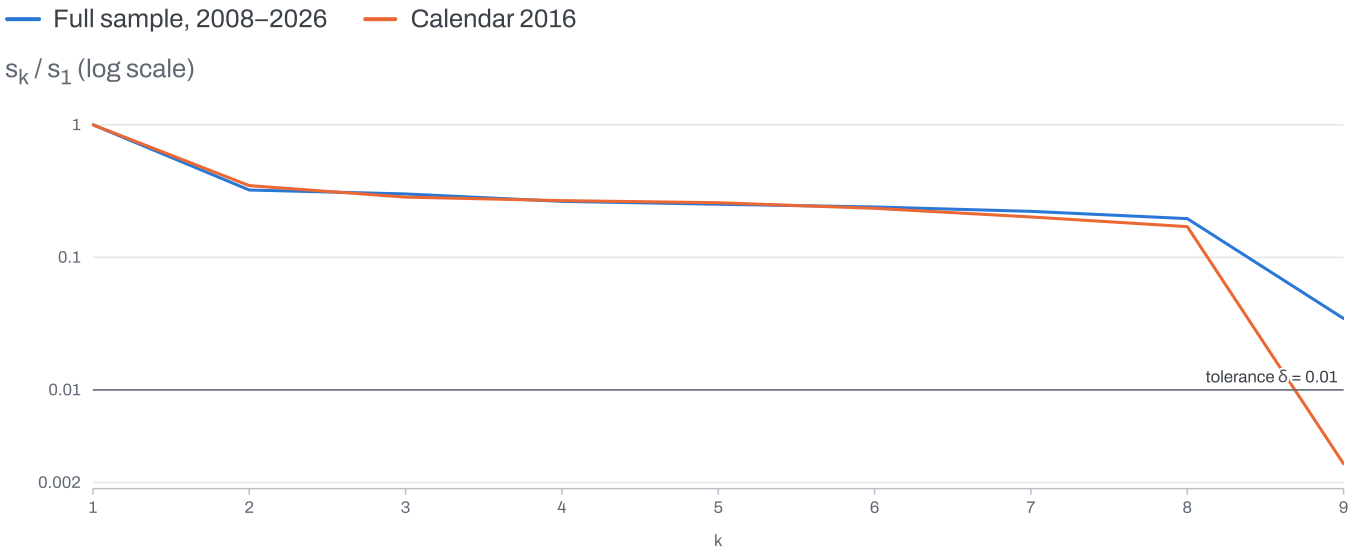
Standardize the daily returns of Electronic and its eight sub-industries (divisor T), and divide the standardized matrix by $\sqrt{T}$. Find its numerical rank at $\delta = 0.1$ and at $\delta = 0.01$, on the full sample and in 2016.

SOLUTION

On the full sample the smallest singular value is $s_9 = 0.0346\, s_1$, and in 2016 it is $0.00277\, s_1$ (Exhibit 8). At $\delta = 0.01$ the numerical rank is therefore 9 on the full sample and 8 in 2016; at $\delta = 0.1$ it is 8 in both. `matrix_rank` reports 9 in every calendar year.

Scale matters too: for the centred returns without standardizing (the covariance scale), the full-sample ratio is 0.0329 instead of 0.0346. A numerical rank needs both its tolerance and its scale.

Exhibit 8: Singular values of Electronic and its eight sub-industries



Note: the ratios s_k/s_1 for the standardized daily returns (divisor T) of Electronic and its eight sub-industries, log scale. The grey line is the tolerance $\delta = 0.01$: 9 ratios of the full sample lie above it, and 8 of 2016.

PITFALL **"Full rank" is not "well determined"**

`np.linalg.matrix_rank` reports 9 for Electronic and its sub-industries in every year, yet in 2016 the smallest singular value is 0.3% of the largest. Coefficients on such nearly dependent columns move a lot when the data move a little (LM11). Report s_N/s_1 or κ next to any rank, and choose δ for the precision of the data, not of the computer.

The calls of this section, on the December 2008 design:

PYTHON

```python
import numpy as np

# X: 22 x 28 sector returns of December 2008; y: TAIEX's 22 returns
design = np.column_stack([np.ones(len(y)), X])       # (1 X), 22 x 29
b_lstsq = np.linalg.lstsq(design, y, rcond=None)[0]   # minimum norm
b_pinv = np.linalg.pinv(design) @ y                   # the same, (12.7)
# s comes in descending order, and the rows of Vh are the v_k
U, s, Vh = np.linalg.svd(design, full_matrices=False)
print(np.linalg.matrix_rank(design))                  # 22
print(np.abs(b_lstsq - b_pinv).max() < 1e-12)         # True
```

導讀 **奇異值分解與數值秩**

奇異值分解（singular value decomposition）把任何矩陣寫成「正交變換（orthogonal transformation，即旋轉或鏡射）、沿座標軸伸縮、再一次正交變換」：可逆（invertible）的 2×2 矩陣把單位圓映成橢圓，奇異值（singular value）就是橢圓的半軸長。把報酬矩陣中心化（centring）並除以 $\sqrt{T}$，對 $\tilde{X}/\sqrt{T}$ 做奇異值分解，樣本共變異數矩陣（sample covariance matrix）的特徵值（eigenvalue）就等於奇異值的平方；特徵值互不相同時，右奇異向量（right singular vector）就是主成分（principal component）方向。正規方程（normal equations）先算 $\tilde{X}^\top\tilde{X}$，會把條件數（condition number）平方，所以軟體直接在 $\tilde{X}$ 上計算。2008 年 12 月只有 22 個報酬，卻有 29 個係數，秩（rank）只有 22，任何反應變數（response）都能完全配適，判定係數（R-squared）等於 1 只是算術結果。擬反矩陣（pseudoinverse）從無限多組解中挑出範數（norm）最小的一組；拿掉半導體一樣完全配適，係數卻全變了。數值秩（numerical rank）取決於容許誤差（tolerance）與尺度（scale），`matrix_rank` 說「滿秩（full rank）」不代表係數估得準。

KNOWLEDGE CHECK 4

A 22×29 design M has rank 22. How many exact solutions do the normal equations have, what is the dimension of the set of solutions, and which solution does `np.linalg.lstsq` return?

ANSWER

Infinitely many: $M^{+}\mathbf{y}$ plus any vector of the null space, a set of dimension $29 - 22 = 7$. `lstsq` returns the solution with the smallest norm (NumPy reference), the vector $M^{+}\mathbf{y}$ that `pinv` also gives.

6 CHOLESKY, WHITENING AND COLOURING

- [ ] compute a Cholesky factor by recursion, and interpret its entries as the slopes and residual standard deviations of sequential regressions

- [ ] whiten and colour returns with a Cholesky factor, and explain why whitened returns have a mean squared length equal to the number of assets

The 2×2 case

LM7 classified a symmetric 2×2 matrix by completing the square, (7.4). Read as a factorization of a 2×2 covariance matrix $\hat{\Sigma}$, the completed square says

$$\begin{pmatrix} \hat{\sigma}_{11} & \hat{\sigma}_{21} \\ \hat{\sigma}_{21} & \hat{\sigma}_{22} \end{pmatrix} = LL^{\top},$$
$$L = \begin{pmatrix} \sqrt{\hat{\sigma}_{11}} & 0 \\ \hat{\sigma}_{21}/\sqrt{\hat{\sigma}_{11}} & \sqrt{\det \hat{\Sigma}/\hat{\sigma}_{11}} \end{pmatrix}.$$

The factor L is lower triangular. Its diagonal is real and positive exactly when $\hat{\sigma}_{11} > 0$ and $\det \hat{\Sigma} > 0$, which is LM7's test for a positive definite matrix.

The factorization

THEOREM **Cholesky factorization**

Let M be a symmetric $N \times N$ matrix. Then M is positive definite if and only if $M = LL^{\top}$ for a lower-triangular L with positive diagonal entries, and that L is unique. Proof: Deep Dive below; see also Strang (2016, §6.5) and Horn and Johnson (2013, ch. 7).

For $M = \hat{\Sigma}$, multiply out $LL^{\top}$. Entry (i, j) with $i \geq j$ is $\sum_{p \leq j} l_{ip} l_{jp} = \hat{\sigma}_{ij}$, and solving for the last term gives the entries column by column:

$$\begin{aligned} l_{jj} &= \Big(\hat{\sigma}_{jj} - \sum_{p<j} l_{jp}^2 \Big)^{1/2}, \\ l_{ij} &= \frac{1}{l_{jj}} \Big(\hat{\sigma}_{ij} - \sum_{p<j} l_{ip} l_{jp} \Big) \quad (i > j). \end{aligned} \tag{12.8}$$

The leading $j \times j$ block of $LL^{\top}$ uses only the leading block of L, so each leading principal minor of $\hat{\Sigma}$ is $l_{11}^2 \cdots l_{jj}^2$. The pivots d_j of LM9 and LM10, ratios of successive leading minors, are therefore $d_j = l_{jj}^2$. If $\hat{\Sigma}$ is only semidefinite, some l_{jj}^2 is zero or, in floating point, tiny or negative. December 2008's $\hat{\Sigma}$, 22 returns on 28 sectors, has rank 21, at most $T - 1$ (LM10), and `np.linalg.cholesky` raises `LinAlgError` on it.

DEEP DIVE **Existence and uniqueness of the Cholesky factor** optional · click to expand

Here $\mathbf{x}$ is a vector of $\mathbb{R}^N$, K the leading block of M of size $N-1$, $\mathbf{a}$ the rest of the last column and m_{NN} the corner entry. The factor of K is L_K, and $\mathbf{l}^\top$ is the last row of L without its corner.

If. When $M = LL^\top$ with L invertible, $\mathbf{x}^\top M\mathbf{x} = \|L^\top\mathbf{x}\|^2 > 0$ for $\mathbf{x} \neq \mathbf{0}$.

Only if, by induction on N. For $N = 1$ the square root of the positive entry is the only choice. For $N > 1$ write

$$M = \begin{pmatrix} K & \mathbf{a} \\ \mathbf{a}^\top & m_{NN} \end{pmatrix}, \qquad L = \begin{pmatrix} L_K & \mathbf{0} \\ \mathbf{l}^\top & l_{NN} \end{pmatrix}.$$

The block K is positive definite, so exactly one L_K exists. The last column requires $\mathbf{l} = L_K^{-1}\mathbf{a}$, and the corner $l_{NN}^2 = m_{NN} - \mathbf{l}^\top\mathbf{l}$. Take $\mathbf{x}$ with $-K^{-1}\mathbf{a}$ above a last entry 1. Positive definiteness then gives

$$\begin{aligned} 0 < \mathbf{x}^\top M\mathbf{x} &= m_{NN} - \mathbf{a}^\top K^{-1}\mathbf{a} \\ &= m_{NN} - \mathbf{l}^\top\mathbf{l}, \end{aligned}$$

so exactly one $l_{NN} > 0$ fits. ■

A chain of regressions

Each column of L has a regression meaning. Define $\mathbf{z}_t := L^{-1}(\mathbf{R}_t - \bar{\mathbf{R}})$, so that $\mathbf{R}_t - \bar{\mathbf{R}} = L\mathbf{z}_t$. The sample covariance of $\mathbf{z}_t$ is $L^{-1}\hat{\Sigma}L^{-\top} = I_N$: its entries are uncorrelated, with variance 1. Row j of $L\mathbf{z}_t$ reads

$$R_{j,t} - \bar{R}_j = \sum_{p<j} l_{jp} z_{p,t} + l_{jj} z_{j,t}.$$

The series $z_{p,t}$ with $p < j$ span the same space as the first $j-1$ centred returns, and $z_{j,t}$ is uncorrelated with them. So the sum is the least-squares fit of asset j on the assets before it, and $l_{jj}z_{j,t}$ is its residual. Each l_{jj} is a residual standard deviation, and l_{11} is the first asset's own.

EXAMPLE 9

Cholesky by hand for two sectors

For Semiconductor and FinancialInsurance on the full sample, $\hat{\sigma}_{11} = 2.323354 \times 10^{-4}$, $\hat{\sigma}_{21} = 1.066603 \times 10^{-4}$ and $\hat{\sigma}_{22} = 1.481729 \times 10^{-4}$ (daily, divisor T). Compute L with (12.8) and interpret its entries.

SOLUTION

Column 1: $l_{11} = \sqrt{2.323354 \times 10^{-4}} = 0.0152426$ and $l_{21} = 1.066603 \times 10^{-4} / 0.0152426 = 0.006998$.
Column 2: $l_{22} = (1.481729 \times 10^{-4} - 0.006998^2)^{1/2} = 0.009960$.

The ratio $l_{21}/l_{11} = \hat{\sigma}_{21}/\hat{\sigma}_{11} = 0.4591$ is the slope of FinancialInsurance on Semiconductor. The entry l_{22} is the residual standard deviation of that regression, 0.996% a day against FinancialInsurance's own 1.217%. Adding Food gives S3, whose $l_{33} = 0.008594$ equals Food's residual standard deviation on the other two. Exhibit 9 gives S5's whole factor.

Exhibit 9: Cholesky factor of S5's covariance matrix

	1	2	3	4	5
1 Semiconductor	1.5243				
2 FinancialInsurance	0.6998	0.9960			
3 Food	0.4336	0.4779	0.8594		
4 Plastics	0.7238	0.5162	0.1831	1.3034	
5 ShippingTransportation	0.6415	0.5995	0.2050	0.1109	1.4358

Note: the lower-triangular L with $\hat{\Sigma} = LL^\top$, S5 in this order, full sample, divisor T, in % a day. Each diagonal entry is the residual standard deviation of its sector on the sectors above it. The squares of a row sum to the sector's variance: Semiconductor's $1.5243^2 = 2.323$ (%²).

Whitening

The vector $\mathbf{z}_t$ is the **whitened** return. It comes from a triangular solve of $L\mathbf{z}_t = \mathbf{R}_t - \bar{\mathbf{R}}$, worked down the rows, not from an inverse (on inverses, see Higham 2002a, ch. 14):

$$\begin{gathered} \mathbf{z}_t = L^{-1}(\mathbf{R}_t - \bar{\mathbf{R}}), \qquad \frac{1}{T}\sum_t \mathbf{z}_t \mathbf{z}_t^\top = I_N, \\ \|\mathbf{z}_t\|^2 = (\mathbf{R}_t - \bar{\mathbf{R}})^\top \hat{\Sigma}^{-1} (\mathbf{R}_t - \bar{\mathbf{R}}). \end{gathered} \tag{12.9}$$

The middle identity is $L^{-1}\hat{\Sigma}L^{-\top} = I_N$, and the last uses $\hat{\Sigma}^{-1} = L^{-\top}L^{-1}$. The same factor solves a system with matrix $\hat{\Sigma}$ by two triangular solves, down the rows of L and then up those of $L^\top$.

Since $\mathbf{z}^\top\mathbf{z} = \operatorname{tr}(\mathbf{z}\mathbf{z}^\top)$ for any vector and the trace is linear,

$$\frac{1}{T}\sum_t \|\mathbf{z}_t\|^2 = \operatorname{tr}\left(\frac{1}{T}\sum_t \mathbf{z}_t\mathbf{z}_t^\top\right) = \operatorname{tr} I_N = N,$$

exactly, in every sample, with divisor T. A day with $\|\mathbf{z}_t\|^2$ far above N is unusual for the joint pattern of its returns, not only for their size. Statisticians call $\|\mathbf{z}_t\|^2$ the squared Mahalanobis distance (LM16). The vector $\mathbf{z}_t$ depends on the order of the assets, because the order changes L; its squared length does not, by (12.9).

EXAMPLE 10

How unusual was 7 April 2025?

On 7 April 2025 the S5 sectors returned −9.95%, −9.89%, −8.66%, −9.90% and −9.73% (Semiconductor, FinancialInsurance, Food, Plastics, ShippingTransportation). Whiten the day with Exhibit 9's factor, and compare it with the other days.

SOLUTION

Subtracting the sample means gives −10.048%, −9.939%, −8.693%, −9.947% and −9.772%. Forward substitution with L in % a day starts with $z_1 = -10.048/1.5243 = -6.592$, then $z_2 = (-9.939 - 0.6998 z_1)/0.9960 = -5.347$, and continues down the rows (Exhibit 10).

The squared length is $\|\mathbf{z}\|^2 = 89.30$, the largest of the 4,382 days, while the mean over all days is exactly 5. In the reversed order the entries change, but the squared length is again 89.30.

Exhibit 10: 7 April 2025 whitened, and the five most unusual days

Sector	Return (%)	$\mathbf{z}$, given order	$\mathbf{z}$, reversed order
Semiconductor	−9.95	−6.592	−1.541
FinancialInsurance	−9.89	−5.347	−2.709
Food	−8.66	−3.815	−5.109
Plastics	−9.90	−1.317	−4.518
ShippingTransportation	−9.73	−0.981	−5.751
Squared length $\|\mathbf{z}\|^2$		89.30	89.30

Rank	Date	$\|\mathbf{z}_t\|^2$	TAIEX (%)
1	2025-04-07	89.30	−9.70
2	2026-07-31	74.13	+7.98
3	2025-04-10	65.98	+9.25
4	2020-07-27	61.28	+2.31
5	2026-07-17	55.58	−6.47

Note: S5 with the full-sample means and $\hat{\Sigma}$ (divisor T); in the reversed order ShippingTransportation comes first, and each row gives its sector's entry. No close on these dates or the dates before them was repaired.

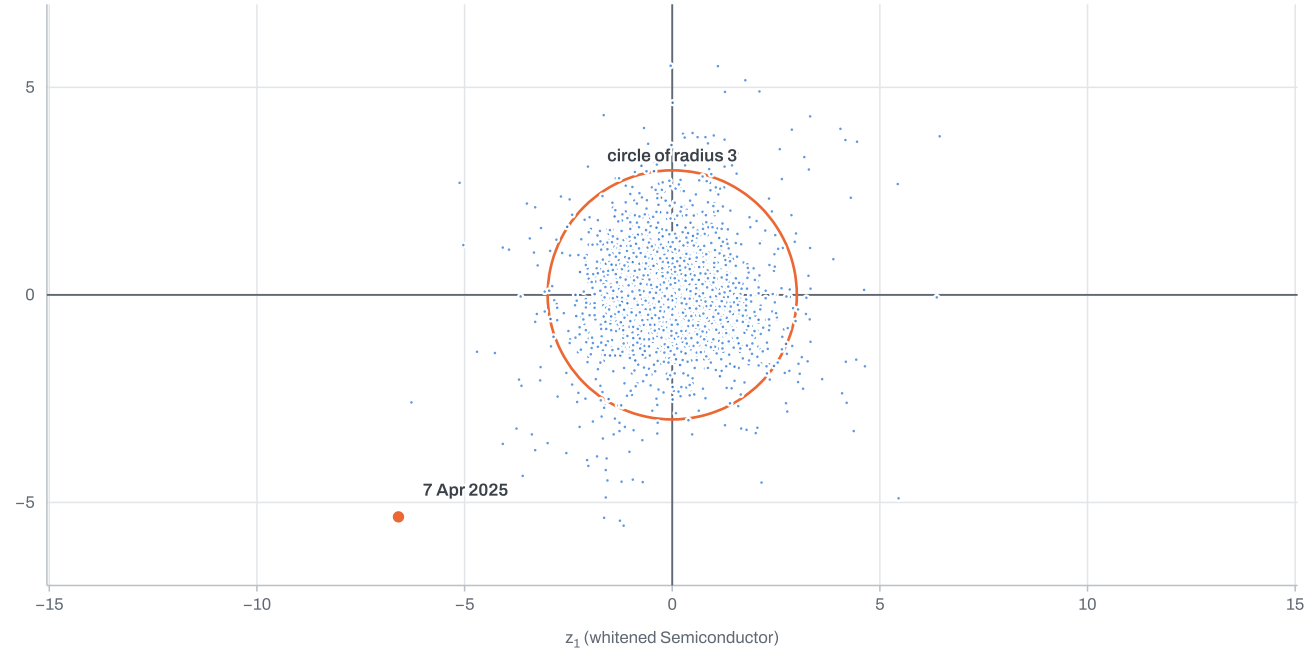
Exhibit 11 shows whitening for the first two sectors: a tilted elliptical cloud becomes a round one, and 7 April 2025 lies far outside both.

Exhibit 11: Whitening two sectors

FinancialInsurance, centred (% a day)



z_2 (whitened FinancialInsurance)



Note: top, the centred daily returns of Semiconductor and FinancialInsurance (% a day), 4,382 days, with the ellipse on which $z_{1,t}^2 + z_{2,t}^2 = 9$, the squared whitened length of the pair. Bottom, the whitened pairs $(z_{1,t}, z_{2,t})$ with the same curve, now a circle of radius 3. 7 April 2025 is marked in both.

PITFALL **A capped day looks less extreme than it was**

On 7 April 2025 four of the five S5 sectors fell between 9.73% and 9.95%. Each stock's daily move is limited to 10% (LM4), so a sector index cannot fall much further: the returns were capped, and 89.30 understates the shock. The day is also the extreme of the sample, so it describes the tail, not a typical day.

Colouring

Run backwards, the factor colours: if vectors $\mathbf{z}$ have identity sample covariance, then $L\mathbf{z}$ has sample covariance $LI_N L^\top = \hat{\Sigma}$. A standard normal draw is a random number with the density φ of LM5. Applied to N independent draws, L gives a simulated vector of centred returns with covariance $\hat{\Sigma}$; LM16 and LM18 develop such simulations.

Whitening in NumPy and SciPy:

PYTHON

```python
import numpy as np
from scipy.linalg import solve_triangular

# X5: T x 5 daily returns of S5, one row per date
Xc = X5 - X5.mean(axis=0)                    # centred returns
Sigma = Xc.T @ Xc / len(Xc)                  # covariance matrix, divisor T
L = np.linalg.cholesky(Sigma)                # lower factor
Z = solve_triangular(L, Xc.T, lower=True).T  # whitened returns, (12.9)
print((Z**2).sum(axis=1).mean())             # 5 up to rounding: N
# scipy.linalg.cholesky(Sigma) returns the upper factor, L transposed
```

IN PYTHON **Companion lab**

The Polars-first notebook `LM12_lab.ipynb` repairs the sector closes, rebuilds the data exhibits and computational examples, and solves the computational practice problems. It ends by colouring standard normal draws with S5's L.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

導讀 **Cholesky 分解：把相關的報酬變成不相關**

對稱矩陣（symmetric matrix）是正定（positive definite）的，若且唯若它能寫成 $LL^\top$，其中 L 是對角線（diagonal）為正的下三角矩陣（lower triangular matrix），而且這個 L 唯一；二階的情形就是 LM7 的配方法（completing the square）。L 的每一行都有迴歸（regression）的意義：l_{jj} 是第 j 個資產對排在前面的資產迴歸後的殘差標準差（residual standard deviation）。用 L 把中心化（centring）報酬解回去，就是白化（whitening）：白化後各分量互不相關（uncorrelated）、變異數（variance）為 1，$\|\mathbf{z}_t\|^2$ 的平均恰好等於資產個數 N。2025 年 4 月 7 日五個類股的 $\|\mathbf{z}\|^2$ 是 89.30，是 4,382 天中最大的一天；資產順序改變時 $\mathbf{z}$ 會變，長度不變，而漲跌幅限制讓這個數字低估了衝擊。反過來用 L 乘上互不相關、變異數為 1 的亂數，就是著色（colouring），可以模擬出共變異數（covariance）為 $\hat{\Sigma}$ 的報酬，留待 LM16 與 LM18 處理。

KNOWLEDGE CHECK 5

For S5 the mean of $\|\mathbf{z}_t\|^2$ over the sample is 5. What is it for the 28 sectors of U28 on the same sample, and why is the answer exact?

ANSWER

The mean is 28. By the trace identity, it equals $\operatorname{tr} I_{28} = 28$ for any sample whose $\hat{\Sigma}$ (divisor T) is positive definite. No model is involved: the identity holds for the data themselves.

SUMMARY

Question	Factorization	Result in this module	NumPy
Which constant mix of sectors best reproduces TAIEX?	$\mathbf{y} = \Pi\mathbf{y} + (I - \Pi)\mathbf{y}$	last 250 returns: R-squared 0.99912, Semiconductor 0.542	`lstsq`, `pinv`
How well conditioned is the data matrix, and what is its rank?	$\tilde{X}/\sqrt{T} = USV^\top$	U28: $\kappa(\tilde{X}) = 10.40$; December 2008: rank 22 of 29	`svd`, `cond`, `matrix_rank`
How unusual is a day, and how are returns simulated?	$\hat{\Sigma} = LL^\top$	7 April 2025: $\|\mathbf{z}\|^2 = 89.30$ for S5	`cholesky`

- Least squares is a projection: the residual is orthogonal to every column, (12.1), and Π is symmetric and idempotent with trace N, (12.2). With an intercept, R-squared is $\cos^2\theta$, (12.4).

- TAIEX on the 28 sector indices of U28 fits the last 250 returns with R-squared 0.99912, because an accounting identity nearly holds. The coefficients are the window's best constant mix; Semiconductor's drifted from 0.189 in 2009 to 0.554 in 2026.
- The SVD follows from the spectral theorem for $M^\top M$. For data $\lambda_k = s_k^2$, (12.5), and the normal equations square the condition number, (12.6).
- With more coefficients than returns, every response fits exactly. The pseudoinverse (12.7) picks the minimum-norm solution; a numerical rank depends on its tolerance and scale.
- A positive definite $\hat{\Sigma}$ has a unique Cholesky factor (12.8), a chain of regressions. Whitened returns (12.9) have identity covariance and mean squared length N.

PRACTICE PROBLEMS

1. Let X be a $T \times N$ matrix of rank $N < T$, and $\Pi = X(X^\top X)^{-1}X^\top$. Which statement is **false**?

   A. The matrix Π is symmetric.

   B. The matrix Π is invertible.

   C. The trace of Π equals N.

2. TAIEX's 22 daily returns of December 2008, regressed on the 28 sector returns with an intercept, give R-squared 1 to machine precision. What does this show?

   A. The coefficients are the market-value weights of the 28 sectors in December 2008.

   B. The same coefficients would reproduce TAIEX's returns in January 2009.

   C. Nothing about TAIEX: a 22×29 design of rank 22 fits any response exactly.

3. Which of these matrices has a Cholesky factor, lower triangular with a positive diagonal?

$$\text{A: } \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix} \quad \text{B: } \begin{pmatrix} 4 & 6 \\ 6 & 9 \end{pmatrix} \quad \text{C: } \begin{pmatrix} 1 & 3 \\ 3 & 4 \end{pmatrix}$$

   A. Matrix A only

   B. Matrices A and B

   C. All three

4. Fit a line with an intercept to the points $(x_t, y_t) = (-1, 0), (0, 1), (1, 1), (2, 4)$. Check that the residuals sum to zero and are orthogonal to $\mathbf{x}$, and compute R-squared in two ways.

5. For $\mathbf{x} = (1, 1, 0)^\top$, compute $\Pi = \mathbf{x}(\mathbf{x}^\top\mathbf{x})^{-1}\mathbf{x}^\top$, check $\Pi^2 = \Pi$ and $\operatorname{tr}\Pi = 1$, and project $\mathbf{y} = (3, 1, 2)^\top$. Is the residual orthogonal to $\mathbf{x}$?

6. Compute by hand the SVD of the matrix below, with the sign rule. Check $|\det M| = s_1 s_2$, and give $\kappa(M)$.

$$M = \begin{pmatrix} 2 & 0 \\ 3 & 2 \end{pmatrix}$$

7. The equations $\beta_1 + 2\beta_2 = 5$ and $2\beta_1 + 4\beta_2 = 10$ have a coefficient matrix M of rank 1 and right-hand side $\mathbf{y} = (5, 10)^\top$. Find all solutions and the minimum-norm solution, and check it against $M^{+}\mathbf{y}$ from (12.7).

8. S5's singular values of $\tilde{X}/\sqrt{T}$ are 0.024240, 0.013487, 0.011470, 0.009351 and 0.006964. Compute the eigenvalues of $\hat{\Sigma}$, $\kappa(\tilde{X})$ and $\kappa(\hat{\Sigma})$, and the digits the normal equations might lose by the rule of thumb.

9. LM7 estimated, for 0050 and 00679B, an annualized covariance matrix with $\sigma_1^2 = 0.040425$, $\sigma_2^2 = 0.018214$ and $\sigma_{12} = -0.002866$. Compute its Cholesky factor with 0050 first, and interpret l_{21}/l_{11} and l_{22}.

10. With the factor of Problem 9, whiten the deviation $(-0.30, 0.10)^\top$: 0050 30 percentage points below its mean and 00679B 10 points above. Then put 00679B first and whiten again. Compare the entries and the squared lengths.

11. *(Python)* Fit TAIEX's last 250 returns on U28 in three ways: `lstsq` on $(\mathbf{1}\ X)$, `solve` on the centred normal equations (12.3), and `pinv`. Report R-squared, the largest gap between the slopes, $\kappa(\tilde{X})$ and $\kappa(\tilde{X}^\top\tilde{X})$.

12. *(Python)* Fit TAIEX on U28 with an intercept in each calendar year from 2009 to 2026, and report the smallest R-squared. For each year, standardize the returns of Electronic and its eight sub-industries within that year (divisor T), and compare their numerical rank at $\delta = 0.01$ with `matrix_rank`.

SOLUTIONS

1. **B** is correct. The columns of Π lie in $\operatorname{col} X$, so Π has rank $N < T$ and is singular: it sends every residual to $\mathbf{0}$. A holds by the transpose rule, and C is (12.2).

2. **C** is correct. With rank 22 equal to the number of rows, the columns span $\mathbb{R}^{22}$, so every response fits exactly. A fails because the exact fit is not unique: Example 7 finds two exact fits that disagree. B is an out-of-sample claim that no in-sample fit can support.

3. **A** is correct. A Cholesky factor exists exactly when the matrix is positive definite. Matrix A has $4 > 0$ and $\det = 20 - 4 = 16 > 0$; its factor has rows $(2, 0)$ and $(1, 2)$. Matrix B is positive semidefinite but singular, $\det = 36 - 36 = 0$, so the recursion gives $l_{22} = 0$, not a positive number. Matrix C is indefinite, $\det = 4 - 9 = -5$, so l_{22}^2 would be negative.

4. The means are 0.5 and 1.5, so $\tilde{\mathbf{x}} = (-1.5, -0.5, 0.5, 1.5)^\top$ and $\tilde{\mathbf{y}} = (-1.5, -0.5, -0.5, 2.5)^\top$. Then $\hat{\beta} = 6/5 = 1.2$ and $\hat{\alpha} = 1.5 - 1.2 \times 0.5 = 0.9$. The fitted values are $(-0.3, 0.9, 2.1, 3.3)^\top$ and the residuals $(0.3, 0.1, -1.1, 0.7)^\top$: they sum to zero, and $\mathbf{x}^\top \hat{\boldsymbol{\varepsilon}} = -0.3 + 0 - 1.1 + 1.4 = 0$. R-squared is $1 - 1.8/9 = 0.8$, and as a squared cosine $6^2/(5 \times 9) = 0.8$.

5. Since $\mathbf{x}^\top \mathbf{x} = 2$, $\Pi = \mathbf{x}\mathbf{x}^\top/2$ has $\frac{1}{2}$ in the four entries of its leading 2×2 block and zeros elsewhere. Multiplying out confirms $\Pi^2 = \Pi$, and the trace is $\frac{1}{2} + \frac{1}{2} = 1$, the number of columns. The projection is $\Pi \mathbf{y} = (2, 2, 0)^\top$, and the residual $(1, -1, 2)^\top$ is orthogonal to $\mathbf{x}$, since $1 - 1 + 0 = 0$.

6. Here $M^\top M$ has rows $(13, 6)$ and $(6, 4)$, trace 17 and determinant 16, so its eigenvalues are 8.5 ± 7.5: $s_1 = 4$ and $s_2 = 1$. The first row of $M^\top M - 16I$ is $(-3, 6)$, so $\mathbf{v}_1 = (2, 1)^\top/\sqrt{5}$. The sign rule turns $(1, -2)^\top/\sqrt{5}$ into $\mathbf{v}_2 = (-1, 2)^\top/\sqrt{5}$. Then $\mathbf{u}_1 = M\mathbf{v}_1/4 = (1, 2)^\top/\sqrt{5}$ and $\mathbf{u}_2 = M\mathbf{v}_2 = (-2, 1)^\top/\sqrt{5}$. Check: $|\det M| = 4 = s_1 s_2$, and $\kappa(M) = 4$.

7. The second equation is twice the first, so the solutions are $(5, 0)$ plus any multiple of the null vector $(-2, 1)$. The one orthogonal to $(-2, 1)$ is $(1, 2)$, with norm $\sqrt{5}$. The matrix $M^\top M$ has eigenvalues 25 and 0, so $s_1 = 5$ and $\mathbf{v}_1 = \mathbf{u}_1 = (1, 2)^\top/\sqrt{5}$. By (12.7), $M^+ = \mathbf{v}_1 \mathbf{u}_1^\top/5$, and $M^+ \mathbf{y} = (1, 2)^\top$.

8. The eigenvalues are $\lambda_k = s_k^2$: 5.876, 1.819, 1.316, 0.874 and 0.485, all times 10^{-4}. Then $\kappa(\tilde{X}) = 0.024240/0.006964 = 3.481$ and $\kappa(\hat{\Sigma}) = 3.481^2 = 12.12$. Since $\log_{10} 12.12 = 1.08$, the normal equations might lose about one digit, and Exhibit 6 finds a gap below 10^{-14}.

9. By (12.8), $l_{11} = \sqrt{0.040425} = 0.20106$, $l_{21} = -0.002866/0.20106 = -0.014254$ and $l_{22} = (0.018214 - 0.014254^2)^{1/2} = 0.134204$. The ratio $l_{21}/l_{11} = \sigma_{12}/\sigma_1^2 = -0.0709$ is the slope of 00679B's returns on 0050's in LM7's sample. The entry l_{22} is 00679B's volatility once the part explained by 0050 is removed: 13.42% against its own 13.50%.

10. With 0050 first, $z_1 = -0.30/0.20106 = -1.492$ and $z_2 = (0.10 - 0.014254 \times 1.492)/0.134204 = 0.587$, so $\|\mathbf{z}\|^2 = 2.57$. With 00679B first, the factor has $l_{11} = \sqrt{0.018214} = 0.134959$, $l_{21} = -0.002866/0.134959 = -0.021236$ and $l_{22} = (0.040425 - 0.021236^2)^{1/2} = 0.199935$. Then $z_1 = 0.10/0.134959 = 0.741$ and $z_2 = (-0.30 + 0.021236 \times 0.741)/0.199935 = -1.422$, and $\|\mathbf{z}\|^2 = 2.57$ again: the entries differ, the squared length does not, as (12.9) requires.

11. R-squared is 0.99912. The three slope vectors agree to better than 10^{-14}, and $\kappa(\tilde{X}) = 17.04$ and $\kappa(\tilde{X}^\top \tilde{X}) = 290.2$, as in Example 6.

12. The smallest yearly R-squared is 0.99800, in 2021. At $\delta = 0.01$ the numerical rank of Electronic and its sub-industries is 9 in 2009, 2010 and 2018 and 8 in the other 15 years. In 2010 the ratio s_9/s_1 is 0.01004, just above the tolerance, so a slightly larger δ would give 8. `matrix_rank` gives 9 in all 18: its default tolerance cannot see that the nine series are nearly dependent.

GLOSSARY

Term	中文	Meaning
Least squares	最小平方法	Choosing coefficients that minimize the sum of squared residuals.
Column space	行空間	All combinations $X\boldsymbol{\beta}$ of the columns of X.
Normal equations	正規方程	$X^\top X\hat{\boldsymbol{\beta}} = X^\top \mathbf{y}$, (12.1): the residual is orthogonal to every column.
Projection matrix	投影矩陣	Π, (12.2): symmetric, idempotent, with trace equal to the number of columns.
Residual	殘差	$\hat{\boldsymbol{\epsilon}} = \mathbf{y} - \hat{\mathbf{y}}$, the part of the response the fit misses.
R-squared	判定係數	Share of the centred response's squared length that a fit with an intercept reproduces, (12.4).
Singular value decomposition	奇異值分解	$M = \sum_{s_k>0} s_k \mathbf{u}_k \mathbf{v}_k^\top$ with orthonormal $\mathbf{u}_k$ and $\mathbf{v}_k$ and singular values $s_1 \geq s_2 \geq \cdots \geq 0$; at full column rank, $M = USV^\top$.
Condition number	條件數	$\kappa(M) = s_1/s_N$; forming $M^\top M$ squares it, (12.6).
Numerical rank	數值秩	Number of singular values above δs_1, for a stated tolerance δ.
Null space	零空間	All vectors that a matrix maps to $\mathbf{0}$.
Pseudoinverse	擬反矩陣	$M^+ = \sum_{s_k>0} \mathbf{v}_k \mathbf{u}_k^\top / s_k$, (12.7); $M^+\mathbf{y}$ is the minimum-norm least-squares solution.
Cholesky factorization	Cholesky 分解	$\hat{\Sigma} = LL^\top$ with L lower triangular with a positive diagonal, (12.8).
Whitening	白化	$\mathbf{z}_t = L^{-1}(\mathbf{R}_t - \bar{\mathbf{R}})$, (12.9): uncorrelated entries with variance 1.
Colouring	著色	$L\mathbf{z}$: vectors with identity covariance turned into vectors with covariance $\hat{\Sigma}$.

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